

Table 1 | Comparison among DeepSeek-V3.2-Base, DeepSeek-V4-Flash-Base, and DeepSeek-V4-Pro-Base. All models are evaluated in our internal framework and share the same evaluation setting. Scores with a gap not exceeding 0.3 are considered to be at the same level. The highest score in each row is in **bold font**, and the second is underlined.

Benchmark (Metric)		# Shots	DeepSeek-V3.2 Base	DeepSeek-V4-Flash Base	DeepSeek-V4-Pro Base
Architecture		-	MoE	MoE	MoE
# Activated Params		-	37B	13B	49B
# Total Params		-	671B	284B	1.6T
World Knowl.	AGIEval (EM)	0-shot	80.1	<u>82.6</u>	83.1
	MMLU (EM)	5-shot	87.8	<u>88.7</u>	90.1
	MMLU-Redux (EM)	5-shot	87.5	<u>89.4</u>	90.8
	MMLU-Pro (EM)	5-shot	65.5	<u>68.3</u>	73.5
	MMMLU (EM)	5-shot	87.9	<u>88.8</u>	90.3
	C-Eval (EM)	5-shot	90.4	<u>92.1</u>	93.1
	CMMLU (EM)	5-shot	88.9	<u>90.4</u>	90.8
	MultiLoKo (EM)	5-shot	38.7	<u>42.2</u>	51.1
	Simple-QA verified (EM)	25-shot	28.3	<u>30.1</u>	55.2
	SuperGPQA (EM)	5-shot	45.0	<u>46.5</u>	53.9
	FACTS Parametric (EM)	25-shot	27.1	<u>33.9</u>	62.6
	TriviaQA (EM)	5-shot	<u>83.3</u>	82.8	85.6
Lang. & Reas.	BBH (EM)	3-shot	87.6	<u>86.9</u>	87.5
	DROP (F1)	1-shot	<u>88.2</u>	88.6	88.7
	HellaSwag (EM)	0-shot	<u>86.4</u>	85.7	88.0
	WinoGrande (EM)	0-shot	<u>78.9</u>	<u>79.5</u>	81.5
	CLUEWSC (EM)	5-shot	<u>83.5</u>	82.2	85.2
Code & Math	BigCodeBench (Pass@1)	3-shot	63.9	56.8	<u>59.2</u>
	HumanEval (Pass@1)	0-shot	62.8	<u>69.5</u>	76.8
	GSM8K (EM)	8-shot	<u>91.1</u>	90.8	92.6
	MATH (EM)	4-shot	<u>60.5</u>	57.4	64.5
	MGSM (EM)	8-shot	<u>81.3</u>	85.7	<u>84.4</u>
	CMath (EM)	3-shot	<u>92.6</u>	93.6	90.9
Long Context	LongBench-V2 (EM)	1-shot	40.2	<u>44.7</u>	51.5

5. Post-Training

5.1. Post-Training Pipeline

Following pre-training, we conducted a post-training phase to yield the final models of DeepSeek-V4 series. Although the training pipeline largely mirrored that of DeepSeek-V3.2, a critical methodological substitution was made: the mixed Reinforcement Learning (RL) stage was entirely replaced by On-Policy Distillation (OPD; Gu et al., 2024; Lu and Lab, 2025).

5.1.1. Specialist Training

The development of domain specialists was conducted by adapting the DeepSeek-V3.2 training pipeline. Specifically, each model was sequentially optimized through an initial fine-tuning phase and subsequent Reinforcement Learning (RL) guided by domain-specific prompts and reward signals. For the RL stage, we implemented the Group Relative Policy Optimization (GRPO) algorithm, maintaining hyper-parameters closely aligned with our prior research (DeepSeek-AI, 2025; DeepSeek-AI, 2025).